



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

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Slot : L23+L24

Course : Exploratory Data Analysis Lab

Experiment 8

Code

```
install.packages("readxl")
```

```
install.packages("cluster")
```

```
library(readxl)
```

```
library(cluster)
```

```
mall <- read_excel("C:\\Users\\vaibh\\Downloads\\Mall Customers.xlsx")
```

```
mall <- as.data.frame(mall)
```

```
str(mall)
```

```
summary(mall)
```

```
data <- mall[, c("Age",  
                "Annual Income (k$)",
```

```
"Spending Score (1-100)"]]
```

```
data <- na.omit(data)
```

```
data_scaled <- scale(data)
```

```
sil_width <- numeric(4)
```

```
for(k in 2:5)
```

```
{
```

```
  km <- kmeans(data_scaled,
```

```
    centers = k,
```

```
    nstart = 25)
```

```
  sil <- silhouette(km$cluster,
```

```
    dist(data_scaled))
```

```
  sil_width[k - 1] <- mean(sil[, 3])
```

```
}
```

```
plot(2:5,
```

```
  sil_width,
```

```
  type = "b",
```

```
  pch = 19,
```

```
  xlab = "Number of Clusters (K)",
```

```
  ylab = "Average Silhouette Width",
```

```
main = "Silhouette Analysis")
```

```
best_k <- (2:5)[which.max(sil_width)]
```

```
cat("Best K =", best_k, "\n")
```

```
kmeanspp <- function(data, k)
```

```
{
```

```
  n <- nrow(data)
```

```
  centers <- matrix(0,
```

```
    nrow = k,
```

```
    ncol = ncol(data))
```

```
  centers[1, ] <- data[sample(1:n, 1), ]
```

```
  for(i in 2:k)
```

```
  {
```

```
    distances <- apply(data, 1, function(x)
```

```
    {
```

```
      min(apply(centers[1:(i-1), , drop = FALSE],
```

```
        1,
```

```
        function(center)
```

```
        {
```

```
          sum((x - center)^2)
```

```
        }))
```

```
)
```

```
probabilities <- distances / sum(distances)
```

```
centers[i, ] <- data[
```

```
  sample(1:n, 1, prob = probabilities),
```

```
]
```

```
}
```

```
return(centers)
```

```
}
```

```
cat("\nRANDOM INITIALIZATION RESULTS\n")
```

```
random_wcss <- numeric(5)
```

```
for(i in 1:5)
```

```
{
```

```
  km_random <- kmeans(data_scaled,
```

```
    centers = best_k,
```

```
    nstart = 1)
```

```
  random_wcss[i] <- km_random$tot.withinss
```

```
  cat("Run", i,
```

```
    "WCSS =", km_random$tot.withinss,
```

```
    "\n")
}

cat("\nK-MEANS++ INITIALIZATION RESULTS\n")

pp_wcss <- numeric(5)

for(i in 1:5)
{
  initial_centers <- kmeanspp(data_scaled, best_k)

  km_pp <- kmeans(data_scaled,
                  centers = initial_centers)

  pp_wcss[i] <- km_pp$tot.withinss

  cat("Run", i,
      "WCSS =", km_pp$tot.withinss,
      "\n")
}

final_centers <- kmeanspp(data_scaled, best_k)

final_km <- kmeans(data_scaled,
                  centers = final_centers)
```

```
cat("\nFINAL CLUSTER CENTERS\n")
```

```
print(final_km$centers)
```

```
cat("\nNUMBER OF CUSTOMERS IN EACH CLUSTER\n")
```

```
print(final_km$size)
```

```
cat("\nFINAL WCSS\n")
```

```
print(final_km$tot.withinss)
```

```
pca <- prcomp(data_scaled)
```

```
plot(pca$x[,1],
```

```
      pca$x[,2],
```

```
      col = final_km$cluster,
```

```
      pch = 19,
```

```
      xlab = "Principal Component 1",
```

```
      ylab = "Principal Component 2",
```

```
      main = "Customer Segmentation using K-Means++")
```

```
legend("topright",
```

```
      legend = paste("Cluster", 1:best_k),
```

```
      col = 1:best_k,
```

```
      pch = 19)
```

```
summary_table <- aggregate(
```

```
  data,
```

```
by = list(Cluster = final_km$cluster),  
FUN = mean  
)
```

```
summary_table$Customers <- as.numeric(  
  table(final_km$cluster)  
)
```

```
names(summary_table) <- c(  
  "Cluster",  
  "Mean_Age",  
  "Mean_Income",  
  "Mean_Spending_Score",  
  "Number_of_Customers"  
)
```

```
summary_table <- summary_table[  
  ,  
  c("Cluster",  
    "Number_of_Customers",  
    "Mean_Age",  
    "Mean_Income",  
    "Mean_Spending_Score")  
]
```

```
cat("\nCLUSTER SUMMARY TABLE\n")
```

```
print(summary_table)
```

Output

